Improved bounds and two exact values for the Zarankiewicz numbers $z(m, n; 3, 3)$

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Abstract

We improve the best known lower bounds for eighteen Zarankiewicz numbers $z(m, n; 3, 3)$, with $13 \le m \le 16$ and $18 \le n \le 23$, by up to 7. The new matrices were found by a local search that treats all cells together and passes each improvement to the neighbouring cells. Combining an elementary deletion argument with two computer-assisted methods, the elimination of column-degree profiles by exactly certified linear programming and an exhaustive orderly generation, we improve nineteen upper bounds and determine $z(13, 20) = 125$ and $z(13, 21) = 131$. We summarize the bounds from the recent literature in this range.

1 Introduction

The Zarankiewicz number $z(m, n; s, t)$ is the maximum number of ones in an $m \times n$ $(0, 1)$-matrix with no $s \times t$ all-ones submatrix [12, 8]. Equivalently, it is the maximum number of edges in a subgraph of $K_{m,n}$ with no $K_{s,t}$ having its s vertices in the part of size m; the matrix is the biadjacency matrix of the graph. We study $s = t = 3$ and write $z(m, n) = z(m, n; 3, 3) = z(n, m)$.

Small values were tabulated by Guy [6], Collins et al. [3] and Tan [11]. General upper bounds include those of Roman [9] and of Davies, Gill and Horsley [4]. Recently, Bhan et al. [2] used an LLM-guided evolutionary search to find lower bounds for $8 \le m \le 16$, $16 \le n \le 23$. They determined three values exactly and left 43 cells open. Some of these were later settled or improved by Afrasyab [1], Hou [7] and Saurabh [10], and in a public repository of computer-certified results [5].

This note builds on these works and makes three contributions.

- *Lower bounds* (Section 3). We improve the lower bound in eighteen cells, by up to 7, with explicit matrices found by a local search.
- *Upper bounds* (Section 4). The density lemma of Collins et al. [3], applied to known exact values, together with profile elimination by exact linear programming (extending [1]) and an orderly generation, improves 19 upper bounds and determines $z(13, 20) = 125$ and $z(13, 21) = 131$. The deletion argument had already been used in [5] to settle three other cells.
- *Summary* (Section 2). Table 1 and Figure 1 give the bounds for all 43 cells with their sources. Some cells listed as open in [2] were already settled in [11, 3].

The matrices, the verifier, the search code and the scripts that produce the tables are available at `https://github.com/joaoamaro70/zarankiewicz`.

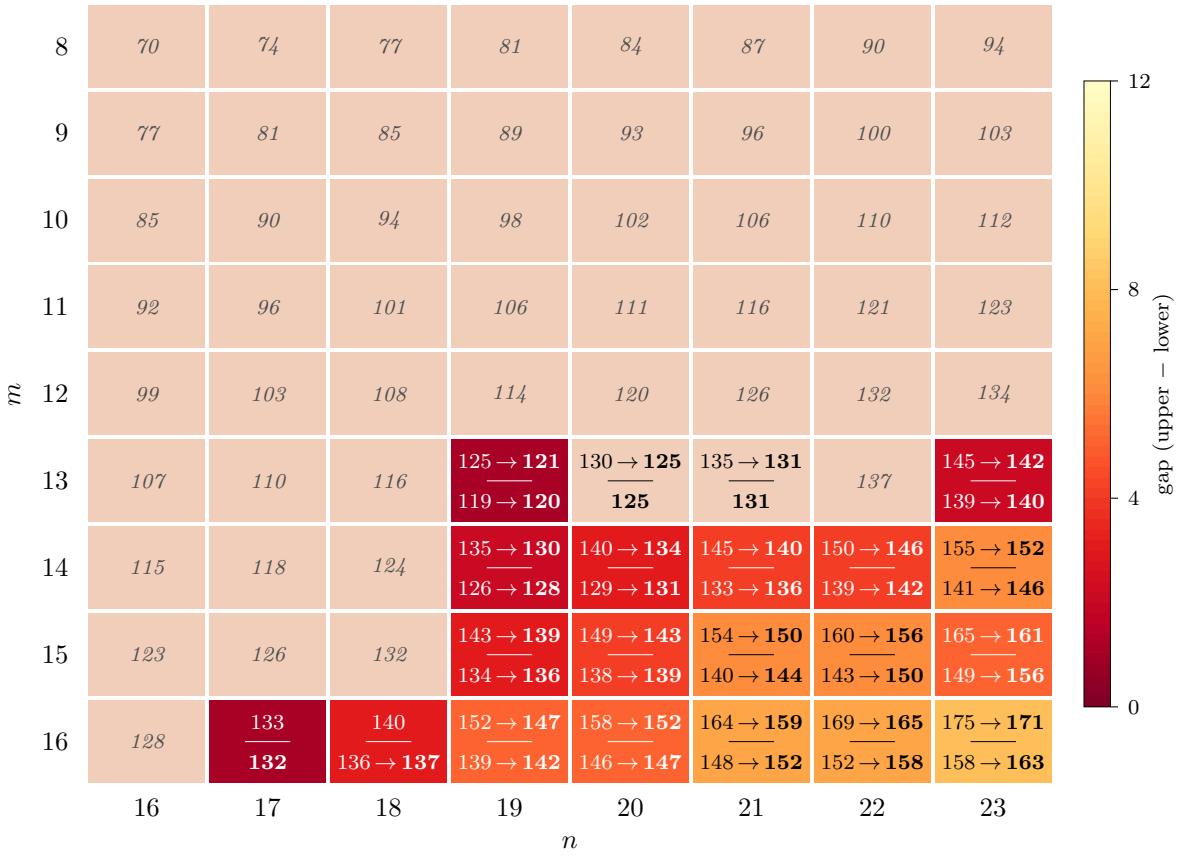


Figure 1: Bounds for $z(m, n; 3, 3)$, in the style of [2, Figure 2]. Beige cells show values that are known exactly. In the other cells, the top number is the best known upper bound, the bottom number is the best lower bound, and the colour shows the gap between them. "$a \to \mathbf{b}$" marks our improvement from the previous best bound a to b, for the upper bound (top) or the lower bound (bottom).

2 Known bounds

We collected the bounds for this range from the following sources.

- Bhan et al. [2]: lower bounds for all cells, together with the upper bounds known at the time. Where Davies, Gill and Horsley [4, Table 2] give an upper bound in this range, it is the one listed there.
- Collins et al. [3, Table 4]: upper bounds for $n \le 18$, several from exhaustive computation, such as $z(13, 17) \le 110$, and the exact value $z(12, 17) = 103$.
- Tan [11, Table 3]: exact values computed with SAT solvers, among them $z(9, 21) = 96$, $z(10, 20) = 102$ and $z(11, 18) = 101$.
- Afrasyab [1]: $z(12, n) = 6n$ for $18 \le n \le 22$ and $z(13, 22) = 137$, with computer-assisted proofs.
- Hou [7]: explicit matrices settling the cells in $\{13, 14, 15\} \times \{17, 18\}$.

- Saurabh [10]: five improved lower bounds.
- The repository [5]: the exact values $z(9,23) = 103$, $z(10,21) = 106$, $z(10,22) = 110$, $z(10,23) = 112$, $z(11,19) = 106$, $z(11,20) = 111$, $z(11,23) = 123$ and $z(12,23) = 134$, with explicit matrices and machine-checked certificates (Lean, DRAT/LRAT and VIPR). Three of them, $z(10,21)$, $z(11,19)$ and $z(11,20)$, follow from the deletion argument of Section 4.

The 2026 preprints [1, 7, 10] and the repository [5] are not peer reviewed. Their lower bounds are explicit matrices that can be checked directly. Their upper bounds rely on computer-assisted proofs that, as their authors note, still await independent review.

Elementary bounds. Two simple facts relate neighbouring cells. Appending a column with two ones creates no 3×3 all-ones submatrix, and deleting a column with the fewest ones loses at most the average number of ones per column (the density lemma of [3, Lemma 3]). This gives the lower bounds

$$z(m,n+1) \geq z(m,n) + 2, \qquad z(m,n) \geq z(m,n+1) - \left\lfloor \frac{z(m,n+1)}{n+1} \right\rfloor, \tag{1}$$

and, read in the other direction, the upper bounds

$$z(m,n) \leq z(m,n+1) - 2, \qquad z(m,n+1) \leq \max\Big\{W : W - \Big\lfloor \frac{W}{n+1} \Big\rfloor \leq z(m,n)\Big\}, \tag{2}$$

together with the same inequalities for rows.

Previous best bounds. For each cell, the *previous lower bound* is the largest published lower bound after applying (1) over the whole range until nothing changes, and the *previous upper bound* is the smallest published upper bound. This already improves several lower bounds of [2]. For example, $z(16,20) \geq 146$ [2] gives $z(16,19) \geq 146 - \lfloor 146/20 \rfloor = 139$, compared with the published 132. Table 1 lists all previous bounds with their sources, next to our results.

Table 1: Bounds for the 43 cells left open in [2]: the previous best bounds with their sources, and the new bounds from this note (in bold). "via (a,b)" means that the bound follows from the bound for (a,b) by (1) or (2). A dagger marks upper bounds that depend on a computer-assisted proof that is not yet peer reviewed [1, 7, 5].

	upper bound			lower bound				
(m,n)	previous	source	new	previous	source	new	gain	status
$(9,23)$	$103^\dagger$	[5]		103	[2]			exact
$(10,20)$	102	[2, Fig. 2]		102	[11]			exact
$(10,21)$	$106^\dagger$	[5]		106	[2]			exact
$(10,22)$	$110^\dagger$	[5]		110	[2]			exact
$(10,23)$	$112^\dagger$	[5]		112	[2]			exact
$(11,18)$	101	[2, Fig. 2]		101	[11]			exact
$(11,19)$	$106^\dagger$	[5]		106	[5]			exact
$(11,20)$	$111^\dagger$	[5]		111	[2]			exact
$(11,23)$	$123^\dagger$	[5]		123	[5]			exact
$(12,17)$	103	[3]		103	[3]			exact
$(12,18)$	$108^\dagger$	[1, 7]		108	[2]			exact
$(12,19)$	$114^\dagger$	[1]		114	[1]			exact
$(12,20)$	$120^\dagger$	[1]		120	[1]			exact
$(12,21)$	$126^\dagger$	[1]		126	[1]			exact
$(12,23)$	$134^\dagger$	[5]		134	[5]			exact
$(13,17)$	110	[3]		110	[7]			exact

	upper bound			lower bound				
(m, n)	previous	source	new	previous	source	new	gain	status
$(13, 18)$	116	[3]		116	[7]			exact
$(13, 19)$	125	[2, Fig. 2]	**121**†	119	via $(13, 20)$	**120**	+1	open
$(13, 20)$	130	[2, Fig. 2]	**125**†	125	via $(13, 21)$			**exact (new)**
$(13, 21)$	135	[2, Fig. 2]	**131**†	131	via $(13, 22)$			**exact (new)**
$(13, 22)$	137†	[1]		137	[2]			exact
$(13, 23)$	145	[2, Fig. 2]	**142**	139	via $(13, 22)$	**140**	+1	open
$(14, 17)$	118	[3]		118	[2]			exact
$(14, 18)$	124	[3]		124	[2]			exact
$(14, 19)$	135	[2, Fig. 2]	**130**†	126	[10]	**128**	+2	open
$(14, 20)$	140	[2, Fig. 2]	**134**†	129	via $(15, 20)$	**131**	+2	open
$(14, 21)$	145	[2, Fig. 2]	**140**†	133	via $(13, 21)$	**136**	+3	open
$(14, 22)$	150	[2, Fig. 2]	**146**†	139	via $(13, 22)$	**142**	+3	open
$(14, 23)$	155	[2, Fig. 2]	**152**	141	via $(13, 23)$	**146**	+5	open
$(15, 17)$	126	[3]		126	[7]			exact
$(15, 18)$	132	[3]		132	[2]			exact
$(15, 19)$	143	[2, Fig. 2]	**139**	134	via $(15, 18)$	**136**	+2	open
$(15, 20)$	149	[2, Fig. 2]	**143**†	138	[2]	**139**	+1	open
$(15, 21)$	154	[2, Fig. 2]	**150**†	140	via $(15, 20)$	**144**	+4	open
$(15, 22)$	160	[2, Fig. 2]	**156**	143	[2]	**150**	+7	open
$(15, 23)$	165	[2, Fig. 2]	**161**	149	[2]	**156**	+7	open
$(16, 17)$	133	[3]		132	[1]			open
$(16, 18)$	140	[3]		136	[10]	**137**	+1	open
$(16, 19)$	152	[2, Fig. 2]	**147**	139	via $(16, 20)$	**142**	+3	open
$(16, 20)$	158	[2, Fig. 2]	**152**†	146	[2]	**147**	+1	open
$(16, 21)$	164	[2, Fig. 2]	**159**†	148	via $(16, 20)$	**152**	+4	open
$(16, 22)$	169	[2, Fig. 2]	**165**	152	via $(16, 23)$	**158**	+6	open
$(16, 23)$	175	[2, Fig. 2]	**171**	158	[2]	**163**	+5	open

3 Lower bounds

A matrix is *valid* if it has no 3×3 all-ones submatrix, that is, if every three rows share at most two columns with ones. We write $|X|$ for the number of ones in X. The search (Algorithm 1) works on all cells of the table at once and keeps every matrix valid at all times.

Two kinds of random move are used. In *rip-up and repair*, we remove a few random ones from X and call Greedy. In *simulated annealing*, we put a one in a random cell and then remove one entry from each 3×3 all-ones submatrix this creates; this move is accepted by the Metropolis rule instead of the test in line 18. In part of the time slices, the same search runs over matrices made of circulant blocks, and the results are completed with Greedy and added to the pool.

None of these ingredients is new. Rip-up and repair is the strategy found by the evolutionary search of Bhan et al. [2], Saurabh [10] used a similar kicked-greedy search, and Hou [7] derived several values from a single core matrix by adding rows and deleting columns. Our only change is to run all cells together and pass every improvement to the neighbouring cells automatically. About three quarters of all improvements first appeared this way, through Improve on a neighbouring cell.

Results. The search improved the lower bound in eighteen cells (column "new" of the lower bounds in Table 1), all with $13 \le m \le 16$ and $18 \le n \le 23$. The matrices are drawn in Appendix A and given as text files in the ancillary files. All runs together took about 7 hours on a single desktop

Algorithm 1 Joint local search over all cells

```
procedure GREEDY(X)
    for each empty cell of X, in random order do
        put a one there if X stays valid
    return X

procedure IMPROVE(c, X)                          ▷ new best matrix X for cell c
    best[c] ← X
    for each neighbour c′ of c do                ▷ (m ± 1, n) and (m, n ± 1)
        Y ← GREEDY(X with a row or column added or deleted)
        if |Y| > |best[c′]| then
            IMPROVE(c′, Y)
        else
            add Y to pool[c′]

while time remains do
    pick a cell c below its upper bound, with probability proportional to 1 + UB(c) − |best[c]|
    X ← GREEDY(best[c], a matrix from pool[c], or the empty matrix)
    for a time slice of 20 seconds do
        Y ← a random move applied to X
        if |Y| ≥ |X|, or rarely if |Y| < |X| then
            X ← Y
        if |X| > |best[c]| then
            IMPROVE(c, X)
```

computer. Every new value was found in the first few minutes, and the remaining hours brought no further improvement.

Other attempts. Three further searches found no matrix better than those in Table 1: SAT on matrices invariant under a cyclic group acting on rows and columns (about 5200 instances), large neighbourhood search in which a window of rows or columns is solved exactly by SAT while the rest of the matrix is fixed (about 75 000 windows proved not to admit an extra one), and a tabu search allowing conflicts (about 42 CPU-hours). In the tabu search, random starts consistently stop exactly at our values. This suggests that these lower bounds are hard to improve with local methods, and possibly optimal in some cells.

Verification. A separate Python script, sharing no code with the search, checks every matrix. It counts the ones and tests all triples of rows and, separately, all triples of columns. It is included in the ancillary files.

4 Upper bounds

We obtained new upper bounds in three ways: an elementary deletion argument, the elimination of column-degree profiles by exact linear programming, and an exhaustive orderly generation. The results of the last two were then propagated with the first. Together they improve 19 upper bounds and settle $z(13, 20) = 125$ and $z(13, 21) = 131$ (column "new" of the upper bounds in Table 1).

4.1 Deletion

We applied (2) to the previous upper bounds and exact values, over the whole range and until nothing changed. The argument is a single deletion step, repeated. For example, a 13×19 matrix with 123 ones has a column with at most $\lfloor 123/19 \rfloor = 6$ ones. Deleting it leaves a 13×18 matrix with at least 117 ones, which contradicts $z(13, 18) = 116$ [3, 7]. Hence $z(13, 19) \leq 122$. In the same way, $z(14, 18) = 124$ [3] gives $z(14, 19) \leq 130$. Applied to the exact values of Tan [11], the same argument settles $z(10, 21) = 106$, $z(11, 19) = 106$ and $z(11, 20) = 111$; these three values had already been obtained in this way in [5].

4.2 Profile elimination by exact linear programming

To show $z(m, n) \leq k - 1$ it suffices to show that no valid matrix has exactly k ones. We enumerate the column-degree profiles $(d_1, \ldots, d_n)$ with $\sum_j d_j = k$ and $\sum_j \binom{d_j}{3} \leq 2\binom{m}{3}$, and eliminate each profile with a linear system that every valid matrix with that profile must satisfy. As in the "block polytope" of Afrasyab [1, Section 3], the variables count columns by their intersection pattern with a few fixed columns, and each constraint is a sum of the conditions "each triple of rows lies in at most two columns"; the pair deficits of [5] are sums of the same conditions. We extend this by fixing up to three columns chosen to be extremal (for example, a pair of columns of the same degree with the largest intersection), splitting into the possible cases, and using branch and bound on the integer variables when the linear relaxation is feasible. Each eliminated case comes with an integer Farkas vector found by a floating-point solver and then checked by a separate program in exact arithmetic, which rebuilds every system from the profile and the case and re-enumerates the profiles; the solver is never trusted.

As validation, the method reproduces $z(10, 22) \leq 110$ [5] (all 4 profiles with 111 ones are eliminated) and eliminates exactly the same 77 of the 83 profiles as [1] for $z(13, 22)$ with 138 ones. In the open cells it proves $z(13, 23) \leq 142$, $z(14, 23) \leq 152$, $z(15, 21) \leq 151$, $z(15, 22) \leq 156$, $z(15, 23) \leq 161$, $z(16, 21) \leq 160$, $z(16, 22) \leq 165$ and $z(16, 23) \leq 171$. These proofs use no values from other sources. Some of these bounds improve further when combined with the orderly generation (Table 1). In the other open cells a few nearly regular profiles survive.

4.3 Orderly generation

Every matrix with at least T ones can be built row by row, adding rows in order of non-increasing degree, so that each prefix is the matrix without its lightest rows. We enumerate these prefixes up to row and column permutations, using canonical forms computed with nauty (canonical augmentation, in the spirit of the exhaustive computations of Collins et al. [3]), and prune with $\sum_j \binom{d_j}{3} \leq 2\binom{m}{3}$ and with upper bounds for $z(j, n)$, $j < m$, on the prefixes. If the enumeration ends empty, $z(m, n) \leq T - 1$. As validation, on eight small cases the number of labelled matrices implied by the enumerated classes and their automorphism groups matches a brute-force count exactly, which checks both completeness and the absence of duplicates.

The enumeration proves $z(13, 19) \leq 121$ (with $T = 122$, about 3 CPU-hours) and $z(13, 20) \leq 125$ (with $T = 128, 127, 126$, about 3 CPU-hours in total). Since the lower bound 125 follows from (1), this gives $z(13, 20) = 125$, and the deletion step then gives $z(13, 21) \leq 131$, so $z(13, 21) = 131$ as well. These proofs use, for pruning, the values $z(j, n)$ for $j \leq 9$ (recomputed and equal to those of [11]), $z(10, n)$ and $z(11, n)$ from [11] and the deletion step, and $z(12, 19) \leq 114$ and $z(12, 20) \leq 120$ from [1].

4.4 Reliability

Each upper bound is only as reliable as the values its proof uses. The bounds marked with a dagger in Table 1 depend on a computer-assisted proof that is not yet peer reviewed: those that come from the orderly generation (through $z(12, 19)$ and $z(12, 20)$ of [1]) or from $z(13, 22) = 137$ [1]. The linear-programming bounds and the deletion steps from [11, 3] have no such dependence. The certificates and the enumeration logs are in the repository.

5 Remarks

Nineteen cells remain open. The largest gap is 8, for $z(16, 23) \in [163, 171]$; the smallest are $z(13, 19) \in [120, 121]$ and $z(16, 17) \in [132, 133]$, followed by $z(13, 23) \in [140, 142]$ and $z(14, 19) \in [128, 130]$. The orderly generation for $z(13, 19)$ with $T = 121$, which would settle that cell, had finished 93 of its 240 parts (all empty) when we stopped it, and in the other cells the profiles that survive the linear programming are few and nearly regular; continuing these computations seems the most direct way to settle more cells. A plain SAT encoding solved by Kissat reproduces the values of [11] up to $z(10, 11) = 64$, but in 1.5 hours it did not find a 16×17 matrix with 132 ones, which we know exists. Splitting by column-degree profiles is much faster on small cases, but for $10 \times n$ some profiles already take more than 30 minutes at $n = 15$. The proofs in [1, 5] instead combine such profile reductions with counting and congruence arguments specific to each cell, and this seems the more promising route.

Acknowledgements

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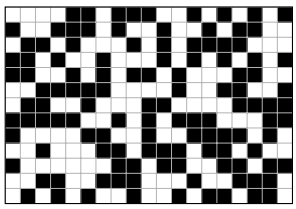
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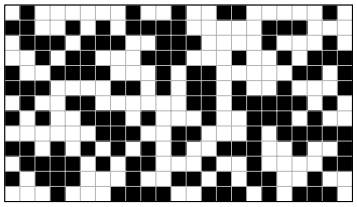
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A The matrices

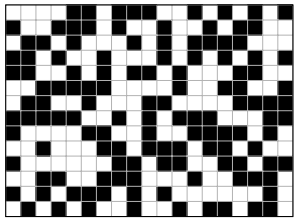
Black cells are ones and white cells are zeros. The same matrices are given as text files in the ancillary files.



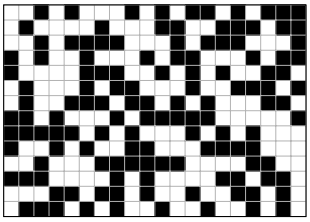
(a) $z(13, 19; 3, 3) \geq 120$



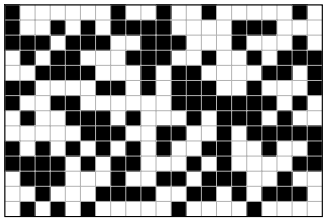
(b) $z(13, 23; 3, 3) \geq 140$



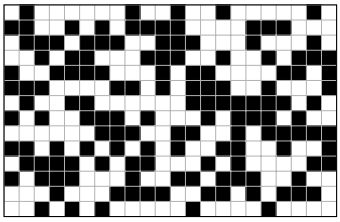
(c) $z(14, 19; 3, 3) \geq 128$



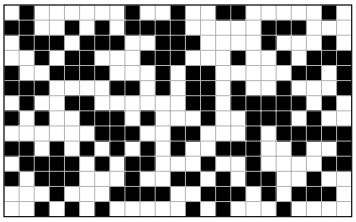
(d) $z(14, 20; 3, 3) \geq 131$



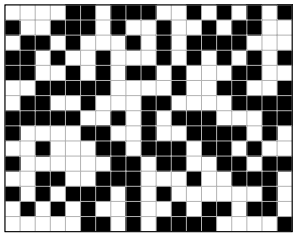
(e) $z(14, 21; 3, 3) \geq 136$



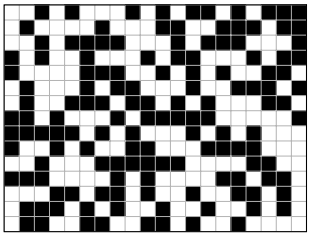
(f) $z(14, 22; 3, 3) \geq 142$



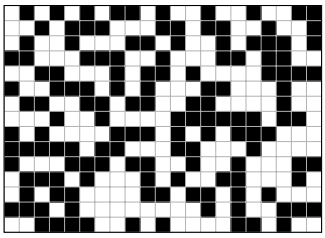
(g) $z(14, 23; 3, 3) \geq 146$



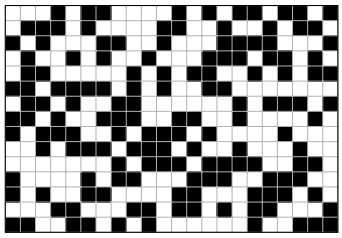
(h) $z(15, 19; 3, 3) \geq 136$



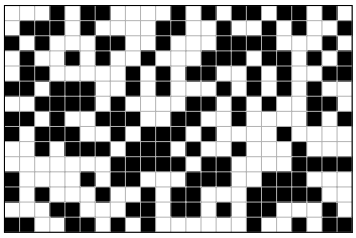
(i) $z(15, 20; 3, 3) \geq 139$



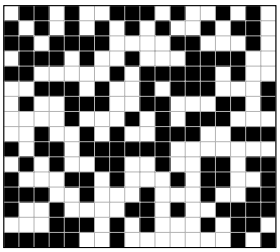
(j) $z(15, 21; 3, 3) \geq 144$



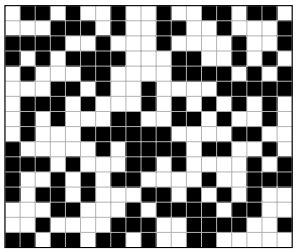
(k) $z(15, 22; 3, 3) \geq 150$



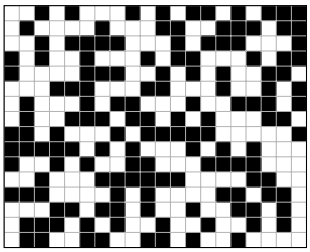
(l) $z(15, 23; 3, 3) \geq 156$



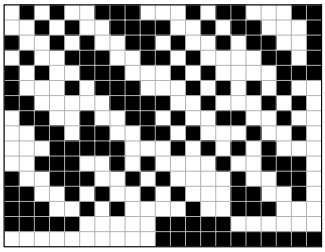
(m) $z(16, 18; 3, 3) \geq 137$



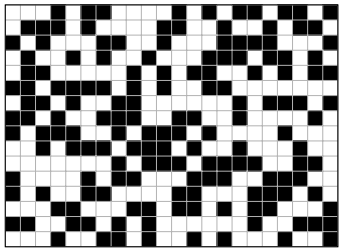
(n) $z(16, 19; 3, 3) \geq 142$



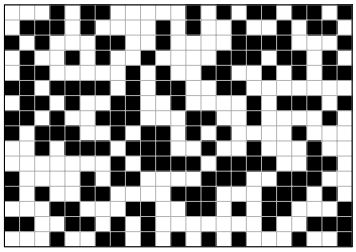
(o) $z(16, 20; 3, 3) \geq 147$



(p) $z(16, 21; 3, 3) \geq 152$



(q) $z(16, 22; 3, 3) \geq 158$



(r) $z(16, 23; 3, 3) \geq 163$